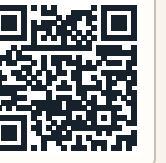


Wandering Black Hole Feeds on Its Gravitational Wake

A 35,000-solar-mass black hole accreting from the dense wake it creates while moving through gas.

arXiv:2608.10719 — Multi-wavelength evidence for off-nuclear intermediate-mass black hole growth.

arXiv:2608.10719 doi:10.48550/arXiv.2608.10719



Background

Intermediate-mass black holes are thought to seed supermassive black holes, but many should wander far from galactic centers. Without a central fuel supply, their growth has remained mysterious.

Direct evidence that wandering black holes can accrete from gravitational wakes has been elusive.

What this paper is selling

This paper reports a 35,000-solar-mass black hole accreting through a gravitational wake. Multi-epoch spectra reveal upstream and downstream gas components, plus absorbers driving rapid broad-line variability.

Key results to remember

- Broad lines, compact continuum, variability, and power-law SED confirm black hole nature.
- Three gas components: blueshifted upstream flow, redshifted dense wake, optically thick absorbers.
- Absorbers within capture radius cause rapidly changing-look broad-line emission.

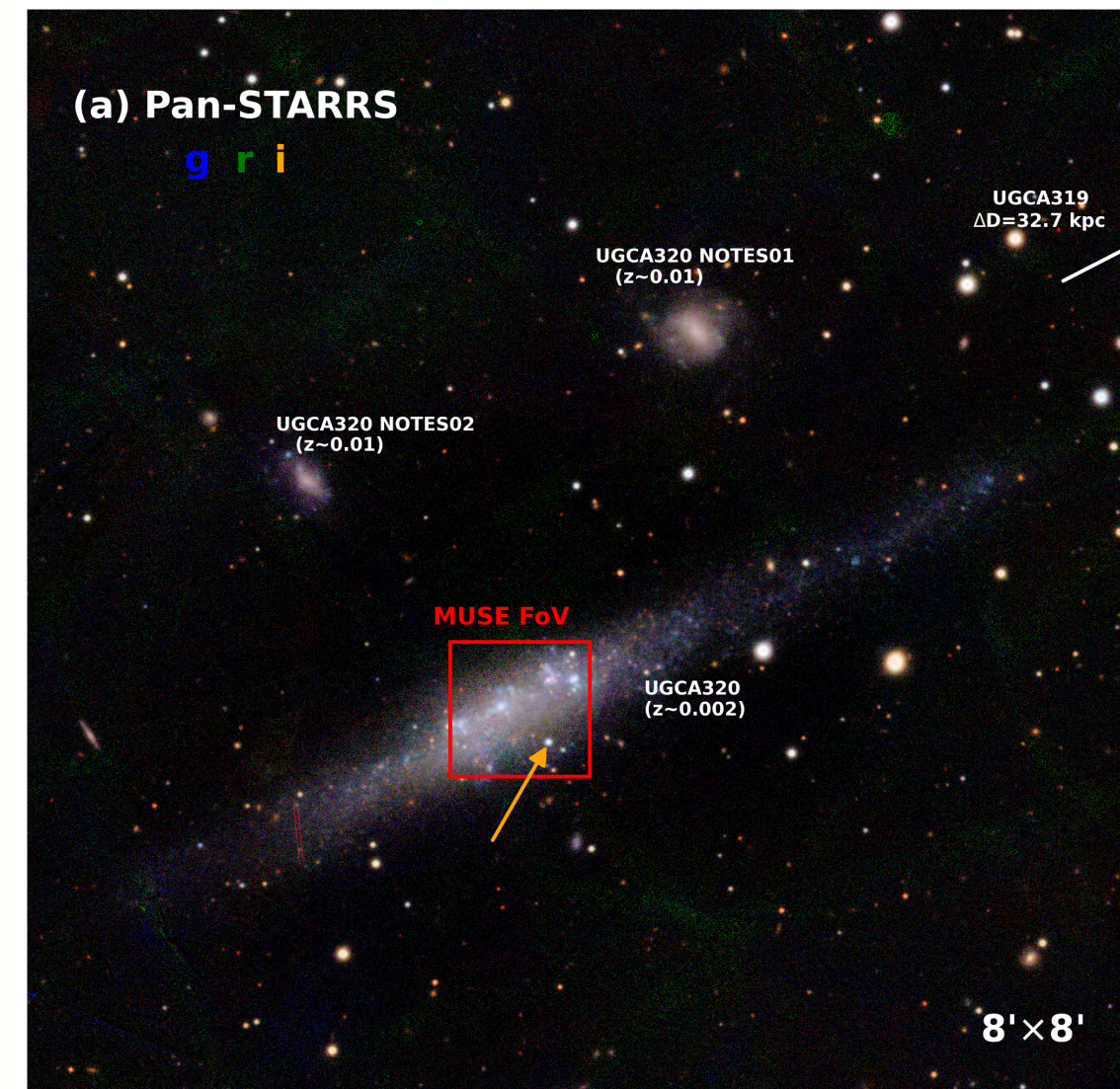


Figure 1 | HST and MUSE views pinpoint the off-nuclear IMBH in UGCA 320, with velocity maps revealing its disturbed gas environment.

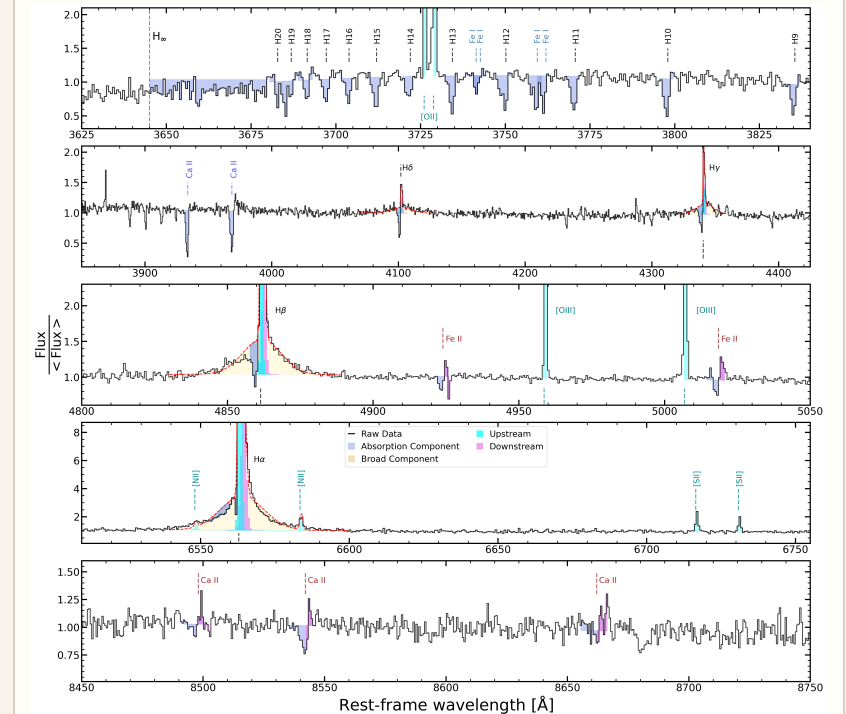


Figure 3 | X-Shooter spectrum disentangles blueshifted absorption, upstream and downstream emission, and broad Balmer lines from the accreting black hole.